

4	A	Based on the given graph, the direction upwards is positive. At time t, the ball is at its maximum height, and its displacement is $+S_1$ (or $+S_2$). At time $2t$, the ball is back at its initial point and is accelerating as it moves downwards (negative velocity). Hence its displacement is zero ($S_1 = S_2$) and the gradient of the displacement-time graph is negative. At time $3t$, the ball hits the ground since its velocity is $-2u$ and acceleration is constant at $-g$. Its displacement is $-S_3$ from the initial point. From $2t$ to $3t$, since the ball is accelerating as it moves downwards, its speed increases. Hence the gradient of the displacement-time graph should be negative with increasing magnitude.
5	C	When B starts to slide: